

- ✓ 1. a) What is preference based approach of consumer's behavior? Explain and interpret the assumptions of preference based approach.
b) Define indifference set. Examine that if a preference relation satisfies strong monotonicity, its indifference curves must be downward sloping.

- ✓ 2. a) Define utility function. Explain the desirability of consumer's preference by the axiom of monotonicity and strong monotonicity.
b) Show that,

$$\text{Strong Monotonicity} \not\Rightarrow \text{Monotonicity} \not\Rightarrow \text{LNS}$$

- ✓ 3. a) What are the twin definitions of convexity? In what respect they are different from strong convexity-examine graphically.
b) Show that.

$$\text{Convexity of preferences} \Leftrightarrow \text{USC}(x) \text{ is convex} \Leftrightarrow u(.) \text{ is quasiconcave}$$

4. a) What are the typical utility functions used in economic applications? Write down their functional forms; also calculate and interpret the notions of MRS.
b) Prove that, if a preference relation satisfies monotonicity and continuity, then there exist a utility function $u(.)$ representing such preference relation.
5. a) Consider a utility maximization problem. If the utility function is continuous and preferences satisfy LNS over the consumption set $X = R_+^N$, then examine the properties satisfied by the Walrasian demand function $x(p, w)$. In what situations the sufficient condition of UMP may be violated.
b) Examine the following properties of preference relations:
- i) Homogeneity
 - ii) Homotheticity

6. Consider the Cobb-Douglas utility function:

$$u(x_1, x_2) = Kx_1^\alpha x_2^{1-\alpha} \quad \text{for some } \alpha \in (0,1) \text{ and } K > 0$$

- i) Derive the Walrasian demand functions and examine their properties.
- ii) How do you get the indirect utility functions? Explain their properties and also verify Roy's identity.
- iii) Solve the expenditure minimization problem and examine the characteristics of the compensated demand function.

7. a) Distinguish between compensating variation (CV) and equivalent variation (EV). In what respect they are different from the consumer's surplus (CS)
- b) Illustrate that for a i) decrease in price of a commodity (when both goods are normal)

$$EV > CS > CV$$

$$\text{ii) for price increase } EV < CS < CV$$

8. a) Define production sets. Graphically explain the properties of the production set.
- b) What is the elasticity of substitution (EoS). Find the elasticity of substitution for the production function $q = f(K, L) = aK + bL$. What will be the EoS if the production function is fixed proportions?

9. a) Define profit function. If the production set Y is closed and have free disposal, then what properties do the profit function satisfies.

b) In case of well-defined utility function, show that $\text{Concavity} \not\Rightarrow \text{Quasi concavity}$

10. Consider a consumer with the utility function: $u(x_1, x_2) = x_1^{\frac{1}{4}} x_2^{\frac{1}{2}}$

- i) Derive the Walrasian and compensated demand functions.

$$\text{ii) Consider a) } \{p^0, m^0\} = \{4, 4, 300\}$$

$$\text{b) } \{p^1, m^0\} = \{2, 4, 300\}$$

Compute the indirect utilities and costs.

- iii) Calculate EV, CV and CS. Based on their relationship examine the nature of the commodities.

502 1st

Course 502

Time 50 minutes

Marks 20

1. (a) Show the impacts of saving ratio in the steady state solution of Solow growth model and prove that exogenous changes in the saving ratio leave only level effects on k and y , but not growth effects.
(b) Show the impacts of changes in the saving ratio on per capita consumption and long run output. Does any steady state solution automatically imply golden rule of capital accumulation? Discuss.
[6+6]
2. Find the elasticity of output per unit of effective labor on the balanced growth path, y^* , with respect to the rate of population growth, n . If $\alpha_k(k^*) = 1/3$, $g = 2\%$, and $\delta = 3\%$, by about how much does a fall in n from 2 percent to 1 percent raise y^* ? [4]
3. Briefly explain classical dichotomy and neutrality of money using the pillars of classical macroeconomics.
[4]

Course 502

Time 50 minutes Marks 20

Answer any one

1. Consider a Ramsey-Cass-Koopmans model with the production function $Y = F(K, AL)$ and real interest rate r ; household's utility function takes the form

$$U = \int e^{-\rho t} u(c(t)) \frac{L(t)}{H} dt$$

and there is CRRA (constant -relative -risk-aversion) utility

$$U(c(t)) = \frac{c(t)^{1-\theta}}{1-\theta}$$

- Construct Household budget constraint and show that there is no-Ponzi-game condition. [5]
 - Solve Euler equation for households' maximization problem. [7]
 - Describe the behavior of the economy in terms of the evolution of c and k . [4]
 - How does the competitive equilibrium maximize the welfare of the representative household. [4]
2. Suppose $Y_t = F(K_t, A_t L_t)$, with having constant returns to scale and the intensive form of the production function satisfying the Inada conditions. Suppose also that $A_{t+1} = (1+g)A_t$, $L_{t+1} = (1+n)L_t$ and the production function is Cobb-Douglas. Assume time is discrete rather than continuous and each individual lives for only two periods.
- Solve the individual's maximization problem where θ determines how much individuals varies in response to differences between r and ρ . [5]
 - Find an expression for k_{t+1} as a function of k_t . [4]
 - Describe how each of the following affects k_{t+1} as a function of k_t :
 - A rise in n .
 - A rise in α . [3]
 - Sketch k as a function of k_t . Does the economy have a balanced growth path? If the initial level of k differs from the value on the balanced growth path, does the economy converge to the balanced growth path? [4]
 - Find the rate of convergence of k to k^* . [4]
3. Consider a simplified version of the models of R&D and growth developed by P. Romer (1990), Gross and Helpman (1991) and Aghion and Howitt (1992) where θ reflects the effect of the existing stock of knowledge on the success of R&D.
- Explain the simplification process to construct the framework. [2]
 - Critically explain the role of a_L (the labor force used in the R&D sector) for the dynamics of knowledge accumulation without considering capital into the model. [6]
 - Assume that $\theta + \beta < 1$ and $n > 0$, and that the economy is on its balanced growth path. Describe how each of the following changes affects the $g_A = 0$ and $g_K = 0$ lines and the position of the economy in (g_A, g_K) space at the moment of the change:
 - An increase in n .
 - An increase in a_K .
 - An increase in θ . [6]
 - Assume that $\beta + \theta = 1$ and $n = 0$,
 - Find the value that A/K must have for g_K and g_A to be equal.
 - Find the growth rate of A and K when $g_K = g_A$.
 - How does an increase in s affect the long-run growth rate of the economy? [6]

Course 502

Time 1 hour

Marks 20

Answer any one

1. Suppose firm i 's output is $Y_i(t) = K_i(t)^\alpha [A(t)L_i(t)]^{1-\alpha}$ and $A(t) = BK(t)$. Here K_i and L_i are the amounts of capital and labor used by firm i and K is the aggregate capital stock. Capital and labor earn their private marginal products. The economy is populated by infinitely lived households that own the economy's capital stock. Individual's discount rate is ρ and their instantaneous utility function is logarithmic. The utility of the representative household takes the constant-relative-risk aversion form and θ is the coefficient of the relative risk aversion. Population growth is zero.
- How the long-run growth is endogenous which is effected by the saving rate according to learning by doing? [4]
 - When output is produced by competitive firms, what will be the demand for the holder of the patent on an idea and the profit? [4]
 - Describe the assumptions to construct the model. [4]
 - When the R&D will be zero and how the long-run growth will be determined? [4]
 - What will be the socially optimal level of L_A ? Explain how it is different from the equilibrium level of L_A . [4]

2. Consider the infinitely lived representative household that derives utility from both consumption (c_t) and leisure ($1-l_t$). The production function of the representative firm is given by

$$f(k_t, l_t, z_t) = z_t k_t^\alpha l_t^{1-\alpha}$$

$$z_{t+1} = \phi z_t + \varepsilon_{t+1}$$

where z_t is the productivity parameter ε is a white noise error term.

- Construct the household's problem and using log utility derive the short-run labor supply curve. Explain the relationship among leisure, real wage, interest rate. [3]
- Prove that long run labor supply is independent of wage when $\phi = \frac{1}{3}$ [3]
- Assuming $\frac{1+r}{1+\rho} = 1$ and functional form of utility of leisure is $v(h) = \frac{\sigma}{1-\sigma} (h)^{\frac{\sigma-1}{\sigma}}$, find the elasticity of labor supply. [4]
- Mention two important limitations of previous business cycle models. Discuss Hansen's indivisible labor supply solution as an attempt to overcome these limitations. [6]
- What are the stylized facts of real business cycle? [4]

Department of Economics; Jahangirnagar University
M. Sc. 2022; 1st Tutorial Examination; Econ- 504 (Econometrics); Time: 1 hour; Marks: 20

Answer **any one** question.

01. A. What do you mean by stochastic process and stationary process? 2/
 B. Explain the AR, MA and ARIMA models with examples. 22
 C. What is the cointegration problem in a time series data? Explain the steps of augmented Engle-Granger (AEG) test for cointegration. 2/
 D. "If the primary objective is forecasting, VAR will do the job." Critically evaluate this statement. 22
 E. Given the DF test result of a variable on time series data. Comment on the result.

Dickey-Fuller test for unit root

		Interpolated Dickey-Fuller		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z (t)	-1.287	-3.696	-2.978	-2.620

MacKinnon approximate p-value for $Z(t)$ = 0.6349

02. A. What do you mean by random walk with drift and without drift model? 2/
 B. Distinguish between time trend stationary (TS) and difference stationary (DS) process. 2/
 C. What is 'unit root' problem in a time series data? Explain the steps of Augmented Dickey-Fuller (ADF) for unit root problem. 2/
 D. What do you mean by the ARCH and GARCH model in time series data forecasting? 22
 E. Given the DF test result of a variable on time series data. Comment on the result.

Dickey-Fuller test for unit root

		----- Interpolated Dickey-Fuller -----		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
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Z(t)	-7.364	-3.702	-2.980	-2.622

MacKinnon approximate p-value for $Z(t)$ = 0.0000

03. A. What is integrated stochastic process? 2/
 B. What is the error correction mechanism (ECM)? What is its relationship with cointegration? 2/
 C. From the BJ methodology explain the identification step to find the value of p, d, q for the model using correlogram and partial correlogram. 22
 D. What are the limitations of the VAR model? 22
 E. Given the result of a variable on time series data. How many lags should be chosen and why?

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-21.7346				.000065	1.71564	1.7754	1.90246
1	126.437	296.34	16	0.000	9.9e-09	-7.09581	-6.79697	-6.16168*
2	148.47	44.066	16	0.000	7.0e-09	-7.49799	-6.96009	-5.81656
3	168.722	40.505	16	0.001	6.3e-09	-7.78148	-7.0045	-5.35274
4	193.517	49.59*	16	0.000	5.0e-09*	-8.36781*	-7.35176*	-5.19176

Department of Economics
Jahangirnagar University
M. Sc. Tutorial Examination
Course: Econometrics (Econ 504)
Time: 1 Hour 10 Minutes Full Marks: 20

- Answer any one from the following questions
 - All parts of a question must be answered consecutively.
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1. A. What is linear probability model (LPM)? What are its limitations? What are the advantages of Probit model over the other models? Explain. 6

B. What is meant by a fixed effect model (FEM)? Since panel data have both time and space dimensions, how does FEM allow for both dimensions? 5

C. What is distributed-lag model and dynamic model? Give examples. 5

D. C. W. J. Granger (1969) has proposed to test the 'causality' between the variables. Explain the procedure using an appropriate economic example. 4

2. A. How would you explain the Logit model? What are the remarkable features of this model? 4

B. Write the Fixed Effect Least-Squares Dummy Variable (LSDV) regression model and explain it. What are the cautions to use of it? 4

B. What are the advantages of panel data over cross-section or time series data? 4

C. How Koyck has proposed an ingenious method of estimating distributed-lag models? 5

D. What is meant by an error components model (ECM)? How does it differ from FEM? 3

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Department of Economics
Jahangirnagar University
M. Sc. 3rd Tutorial Examination
Course: Econometrics (Econ-504)
Time: 1 Hour 10 Minutes; Marks: 20

Answer any one question. All parts of a question try to answer consecutively

- ① A. What is structural or behavioral and reduced-form equation system? Define those with examples. 4
 B. What do you mean by an 'instrumental variable'? Using a simultaneous model, show the estimation procedure of 2SLS method? 4
 C. The problem of simultaneity does not arise in a recursive simultaneous equation model- state whether statement is true or false with reasoning. 4
 D. Verify the following equation whether they are identified by both identification rules or not? 8

Equation no.	1	Y_1	Y_2	Y_3	Y_4	X_1	X_2	X_3
1	$-\beta_{10}$	1	$-\beta_{12}$	$-\beta_{13}$	0	$-\gamma_{11}$	0	0
2	$-\beta_{20}$	0	1	$-\beta_{23}$	0	$-\gamma_{21}$	$-\gamma_{22}$	0
3	$-\beta_{30}$	$-\beta_{31}$	0	1	0	$-\gamma_{31}$	$-\gamma_{32}$	0
4	$-\beta_{40}$	$-\beta_{41}$	$-\beta_{42}$	0	1	0	0	$-\gamma_{43}$

2. A. The method of OLS is not applicable to estimate a structural equation in a simultaneous-equation model- state whether statement is true or false. Explain the reasoning. 6
 B. How could you test the "simultaneity" of a model according to the Hausman specification test? Explain the necessary steps. 3
 C. Explain the steps of the Indirect Least Square (ILS) methods to estimate the simultaneous equation with example. 4
 D. What is the necessary condition for use the 2SLS? Explain the steps of it with example. 4
 E. Given the two regression model's results:

2SLS Regression Results:

$\hat{Y}_{2t} = -4236.89 + 0.5785\hat{Y}_{1t} + 1.8700X_{1t} + 2.0343X_{2t}$					
$se = (125.352)$	$(.0214)$	(0.756)	(0.754)		
$t = (?)$	$(?)$	$(?)$	$(?)$	$R^2 = 0.9776$	

OLS Regression Results:

$\hat{Y}_{2t} = -4378.24 + 0.5776Y_{1t} + 1.8679X_{1t} + 2.0336X_{2t}$					
$se = (135.352)$	$(.0209)$	(0.4571)	(0.6874)		
$t = (?)$	$(?)$	$(?)$	$(?)$	$R^2 = 0.9780$	

Here, Y_1 = GDP; X_1 = private investment; X_2 = government spending; and Y_2 = money supply

"Since the OLS and 2SLS results are practically identical, 2SLS is meaningless." Comment. 3

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507 1st + 2nd

Department of Economics

Jahangirnagar University

1st and 2nd Tutorial (M. Sc. Final Examination 2022)

Course Name: Health Economics

Course No. Econ. 507

Marks: 20+20 Time: 40 Minutes + 40 Minutes

Tutorial 1: Answer any one

1. What is the definition of Health Economics? Give a bird eye view of Health Economics as a separate discipline. How are supply and demand tools applied in the healthcare sector?
2. What are the public interventions in health sector that impacts private decisions? Explain. How cost of health care services became major challenges in the HC sector? How to overcome these challenges?

Tutorial 2: Answer any one

1. Why do some employers choose not to provide insurance, or find themselves unable to do so? How does Mandated Coverage exacerbate this phenomenon? *working uninsured + mandated work*
2. How the tax system influences health insurance demand? Explain how health insurance may affect labour supply related to retirement age and job mobility?

Department of Economics
Jahangirnagar University
3rd Tutorial (M. Sc. Final Examination 2022)
Course Name: Health Economics
Course Number : Econ. 507
Full Marks: 20 Time: 45 Minutes

Answer any one(1) question:

1

[6+14]

- a. What is Clinical Decision Analysis (CDA)? What are the steps of CDA?
- b. Consider a patient who is 56 years old with "Heartburn" for several years. He may have coronary risk factors and a 6-week history of squeezing chest pain. In his **positive** physical examination physicians assigned **80%** coronary artery stenosis (prior probability). Sensitivity and specificity of the Exercise-ECG are **95% and 85%** respectively.
 - i) Calculate posterior probabilities using Bayes Theorem.
 - ii) Perform steps for CDA
 - iii) Graphically present sensitivity analysis based on two pre-test likelihoods.

2

[4+12+4]

- a. Elucidate the concept of QALY as a measure of health benefits using Health Related Quality of Life (HRQoL) indicator.
- b. Describe approaches in obtaining utility values for health states in Cost Utility Analysis (CUA)? Critically demonstrate Rating Scale (RS), Time-Trade Off (TTO) and Standard Gamble (SG) methods for assessing preferences of respondents.
- c. Graphically explain relationship of cardinal utility with TTO method and SG method.

3

[20]

Briefly explain five-phase implementation of DHIS2 to transform the health information system in Bangladesh.

510 1st

ECON 510 [Labor Economics]

Department of Economics

Jahangirnagar University

Time: 1 hour

Tutorial: 01

Total: 20

1. Identify the labor market forces; explain those with the help of diagrams. (5)
2. How can a labor reach his/her best efficient time allocation point about the combination of the labor working hours and the leisure hours? (10)
3. Explain why the female participation rate increased in the labor force overtime. (5)

510 2nd

ECON 510 [Labor Economics]

Department of Economics

Jahangirnagar University

Time: 1 hour

Tutorial: 02

Total: 20

1. What are the various issues in the employee's selection? Explain (5)
2. Briefly discusses the differences between General on- the- job training and Specific on- the- job training. (5)
3. Should an individual attend a four-year University or College? Give your opinion according to the two investment decision rules. (10)